

Calculus I

Complete Notes

Limits · Differentiation · Integration · Applications

DEPARTMENT

Mathematics

SALU, Khairpur

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SESSION

2026

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1. Functions

Definition — Function

A **function** f from set A to set B is a rule that assigns to each element $x \in A$ exactly one element $f(x) \in B$.

$$f : A \longrightarrow B$$

- **Domain:** Set of all valid inputs x
- **Range:** Set of all actual outputs $f(x)$
- **Codomain:** The target set B

Types of Functions

- **Even function:** $f(-x) = f(x)$ (symmetric about y -axis) e.g. x^2 , $\cos x$
- **Odd function:** $f(-x) = -f(x)$ (symmetric about origin) e.g. x^3 , $\sin x$
- **One-to-one (Injective):** Different inputs give different outputs
- **Onto (Surjective):** Every element of B is hit by some x
- **Bijjective:** Both one-to-one and onto $\Rightarrow$ inverse exists

Composite and Inverse Functions

Key Formula

$$(f \circ g)(x) = f(g(x))$$

$$\text{If } y = f(x) \text{ then } x = f^{-1}(y), \quad \text{Domain}(f^{-1}) = \text{Range}(f)$$

Worked Example

Let $f(x) = 2x + 1$ and $g(x) = x^2$. Find $(f \circ g)(x)$.

$$(f \circ g)(x) = f(g(x)) = f(x^2) = 2x^2 + 1$$

2. Limits

Definition — Limit

$\lim_{x \rightarrow a} f(x) = L$ means $f(x)$ gets arbitrarily close to L as x approaches a (but $x \neq a$).

Limit Laws

Key Formula

$$\lim[f(x) \pm g(x)] = \lim f(x) \pm \lim g(x)$$

$$\lim[f(x) \cdot g(x)] = \lim f(x) \cdot \lim g(x)$$

$$\lim \frac{f(x)}{g(x)} = \frac{\lim f(x)}{\lim g(x)}, \quad \lim g(x) \neq 0$$

$$\lim[c \cdot f(x)] = c \cdot \lim f(x)$$

Standard Limits (Must Memorize)

Key Formula

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \quad \lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0 \quad \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \ln a \quad \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e \quad \lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$$

One-Sided Limits

- **Left-hand limit (LHL):** $\lim_{x \rightarrow a^-} f(x)$ — approach from left
- **Right-hand limit (RHL):** $\lim_{x \rightarrow a^+} f(x)$ — approach from right

The limit $\lim_{x \rightarrow a} f(x)$ exists **if and only if** $\text{LHL} = \text{RHL} = L$.

Limits at Infinity

$$\lim_{x \rightarrow \infty} \frac{1}{x^n} = 0 \quad (n > 0), \quad \lim_{x \rightarrow \infty} c = c$$

For rational functions $\frac{p(x)}{q(x)}$ as $x \rightarrow \infty$: divide numerator and denominator by the highest power of x in the denominator.

L'Hôpital's Rule

Theorem — L'Hôpital's Rule

If $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ gives the indeterminate form $\frac{0}{0}$ or $\frac{\infty}{\infty}$, then:

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

provided the right-hand limit exists.

Worked Example

$\lim_{x \rightarrow 0} \frac{\sin x}{x} \rightarrow \frac{0}{0}$ form. Apply L'Hôpital:

$$\lim_{x \rightarrow 0} \frac{\cos x}{1} = \cos 0 = 1$$

3. Continuity

Definition — Continuity at a Point

$f(x)$ is **continuous** at $x = a$ if all three conditions hold:

1. $f(a)$ is defined
2. $\lim_{x \rightarrow a} f(x)$ exists
3. $\lim_{x \rightarrow a} f(x) = f(a)$

Types of Discontinuity

- **Removable:** Limit exists but $\neq f(a)$ — a hole in the graph
- **Jump:** LHL $\neq$ RHL
- **Infinite:** Limit is $\pm\infty$ — vertical asymptote

Theorem — Intermediate Value Theorem

If f is continuous on $[a, b]$ and k is any value between $f(a)$ and $f(b)$, then there exists $c \in (a, b)$ such that $f(c) = k$.

4. Differentiation

Definition — Derivative from First Principles

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Basic Differentiation Rules

Function $f(x)$	Derivative $f'(x)$
c (constant)	0
x^n	nx^{n-1}
e^x	e^x
a^x	$a^x \ln a$
$\ln x$	$\frac{1}{x}$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\cot x$	$-\csc^2 x$
$\sec x$	$\sec x \tan x$
$\csc x$	$-\csc x \cot x$
$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}$
$\cos^{-1} x$	$\frac{-1}{\sqrt{1-x^2}}$
$\tan^{-1} x$	$\frac{1}{1+x^2}$

Product Rule**Key Formula**

$$\frac{d}{dx} [f(x) \cdot g(x)] = f'(x) g(x) + f(x) g'(x)$$

Quotient Rule**Key Formula**

$$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{f'(x) g(x) - f(x) g'(x)}{[g(x)]^2}$$

Chain Rule

Key Formula

$$\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$$

Chain Rule Example

$$y = \sin(x^2)$$

$$\frac{dy}{dx} = \cos(x^2) \cdot 2x = 2x \cos(x^2)$$

Implicit Differentiation

Differentiate both sides w.r.t. x , treating y as a function of x .

Implicit Differentiation Example

$$x^2 + y^2 = 25$$

$$2x + 2y \frac{dy}{dx} = 0 \implies \frac{dy}{dx} = -\frac{x}{y}$$

Logarithmic Differentiation

Take $\ln$ of both sides first. Useful when $y = [f(x)]^{g(x)}$.

Logarithmic Differentiation Example

$$y = x^x$$

$$\ln y = x \ln x \implies \frac{1}{y} \frac{dy}{dx} = \ln x + 1 \implies \frac{dy}{dx} = x^x (\ln x + 1)$$

5. Mean Value Theorems

Theorem — Rolle's Theorem

If f is continuous on $[a, b]$, differentiable on (a, b) , and $f(a) = f(b)$, then $\exists c \in (a, b)$ such that:

$$f'(c) = 0$$

Theorem — Mean Value Theorem (MVT)

If f is continuous on $[a, b]$ and differentiable on (a, b) , then $\exists c \in (a, b)$ such that:

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Theorem — Cauchy's Mean Value Theorem

If f and g are continuous on $[a, b]$ and differentiable on (a, b) , then $\exists c \in (a, b)$ such that:

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

6. Applications of Derivatives

Increasing / Decreasing Functions

- $f'(x) > 0$ on $(a, b) \Rightarrow f$ is **increasing** on (a, b)
- $f'(x) < 0$ on $(a, b) \Rightarrow f$ is **decreasing** on (a, b)
- $f'(c) = 0 \Rightarrow$ **critical point** (possible max or min)

First Derivative Test

- f' changes $+$ to $-$ at $c \Rightarrow$ **local maximum**
- f' changes $-$ to $+$ at $c \Rightarrow$ **local minimum**

Second Derivative Test

Key Formula

At a critical point c where $f'(c) = 0$:

$$f''(c) > 0 \Rightarrow \text{local minimum} \quad f''(c) < 0 \Rightarrow \text{local maximum} \quad f''(c) = 0 \Rightarrow \text{inconclusive}$$

Concavity and Inflection Points

- $f''(x) > 0 \Rightarrow$ concave **up** (cup shape)
- $f''(x) < 0 \Rightarrow$ concave **down** (cap shape)
- $f''(x) = 0$ and changes sign $\Rightarrow$ **inflection point**

Taylor's and Maclaurin's Series

Key Formula

Taylor's Series about $x = a$:

$$f(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \frac{f'''(a)}{3!}(x - a)^3 + \dots$$

Maclaurin's Series (set $a = 0$):

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots$$

Common Maclaurin Series

Key Formula

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots \quad (|x| \leq 1)$$

7. Integration

Definition — Indefinite Integral

$$\int f(x) dx = F(x) + C \quad \text{where } F'(x) = f(x)$$

C is the constant of integration.

Basic Integration Formulas

Function $f(x)$	$\int f(x) dx$
$x^n \ (n \neq -1)$	$\frac{x^{n+1}}{n+1} + C$
$\frac{1}{x}$	$\ln x + C$
e^x	$e^x + C$
a^x	$\frac{a^x}{\ln a} + C$
$\sin x$	$-\cos x + C$
$\cos x$	$\sin x + C$
$\sec^2 x$	$\tan x + C$
$\csc^2 x$	$-\cot x + C$
$\sec x \tan x$	$\sec x + C$
$\frac{1}{\sqrt{1-x^2}}$	$\sin^{-1} x + C$
$\frac{1}{1+x^2}$	$\tan^{-1} x + C$

Techniques of Integration

1. Substitution (u-substitution)

Key Formula

Let $u = g(x)$, then $du = g'(x) dx$:

$$\int f(g(x)) g'(x) dx = \int f(u) du$$

Substitution Example

$\int 2x \cos(x^2) dx$. Let $u = x^2 \Rightarrow du = 2x dx$

$$= \int \cos u du = \sin u + C = \sin(x^2) + C$$

2. Integration by Parts

Key Formula

$$\int u \, dv = uv - \int v \, du$$

LIATE Rule for choosing u : **L**ogarithm, **I**nverse trig, **A**lgebraic, **T**rig, **E**xponential

Integration by Parts Example

$$\int x e^x \, dx. \text{ Let } u = x, \, dv = e^x \, dx \Rightarrow du = dx, \, v = e^x$$

$$= x e^x - \int e^x \, dx = x e^x - e^x + C = e^x(x - 1) + C$$

3. Partial Fractions

Used for rational functions $\frac{P(x)}{Q(x)}$ where $\deg P < \deg Q$.

Partial Fractions Example

$$\int \frac{1}{x^2 - 1} \, dx = \int \left[\frac{1/2}{x - 1} - \frac{1/2}{x + 1} \right] \, dx = \frac{1}{2} \ln |x - 1| - \frac{1}{2} \ln |x + 1| + C$$

Definite Integral

Theorem — Fundamental Theorem of Calculus

Part 1: If $F(x) = \int_a^x f(t) \, dt$, then $F'(x) = f(x)$

Part 2: $\int_a^b f(x) \, dx = F(b) - F(a)$

Properties of Definite Integrals

Key Formula

$$\int_a^a f(x) dx = 0$$

$$\int_a^b f(x) dx = - \int_b^a f(x) dx$$

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

$$\int_{-a}^a f(x) dx = 0 \quad \text{if } f \text{ is odd}$$

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx \quad \text{if } f \text{ is even}$$

8. Applications of Integration

Area Between Curves

Key Formula

$$A = \int_a^b [f(x) - g(x)] dx \quad \text{where } f(x) \geq g(x)$$

Volume of Revolution

Disk Method (rotating $y = f(x)$ about x -axis):

Key Formula

$$V = \pi \int_a^b [f(x)]^2 dx$$

Washer Method (hollow solid):

Key Formula

$$V = \pi \int_a^b \left\{ [f(x)]^2 - [g(x)]^2 \right\} dx$$

Arc Length

Key Formula

$$L = \int_a^b \sqrt{1 + [f'(x)]^2} dx$$

Quick Reference Cheat Sheet

CALCULUS I — MASTER FORMULA SHEET

Everything you need to know for the exam in one place

Derivatives

$$\begin{aligned} \frac{d}{dx}(x^n) &= nx^{n-1} \\ \frac{d}{dx}(e^x) &= e^x \\ \frac{d}{dx}(a^x) &= a^x \ln a \\ \frac{d}{dx}(\ln x) &= \frac{1}{x} \\ \frac{d}{dx}(\sin x) &= \cos x \\ \frac{d}{dx}(\cos x) &= -\sin x \\ \frac{d}{dx}(\tan x) &= \sec^2 x \\ \frac{d}{dx}(\cot x) &= -\csc^2 x \\ \frac{d}{dx}(\sec x) &= \sec x \tan x \\ \frac{d}{dx}(\sin^{-1} x) &= \frac{1}{\sqrt{1-x^2}} \\ \frac{d}{dx}(\cos^{-1} x) &= \frac{-1}{\sqrt{1-x^2}} \\ \frac{d}{dx}(\tan^{-1} x) &= \frac{1}{1+x^2} \end{aligned}$$

Integrals

$$\begin{aligned} \int x^n dx &= \frac{x^{n+1}}{n+1} + C \\ \int \frac{1}{x} dx &= \ln|x| + C \\ \int e^x dx &= e^x + C \\ \int a^x dx &= \frac{a^x}{\ln a} + C \\ \int \sin x dx &= -\cos x + C \\ \int \cos x dx &= \sin x + C \\ \int \sec^2 x dx &= \tan x + C \\ \int \csc^2 x dx &= -\cot x + C \\ \int \sec x \tan x dx &= \sec x + C \\ \int \frac{1}{\sqrt{1-x^2}} dx &= \sin^{-1} x + C \\ \int \frac{1}{1+x^2} dx &= \tan^{-1} x + C \end{aligned}$$

Standard Limits

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \quad \lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0 \quad \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1 \quad \lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \ln a$$

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e \quad \lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1 \quad \lim_{x \rightarrow 0} \frac{\tan x}{x} = 1 \quad \lim_{x \rightarrow \infty} \frac{1}{x^n} = 0$$

Differentiation Rules**Product Rule:**

$$(fg)' = f'g + fg'$$

Quotient Rule:

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

Chain Rule:

$$\frac{d}{dx}f(g(x)) = f'(g(x)) \cdot g'(x)$$

L'Hôpital (0/0 or ∞/∞):

$$\lim \frac{f}{g} = \lim \frac{f'}{g'}$$

Integration Techniques**Substitution:**

$$\int f(g(x))g'(x) dx = \int f(u) du$$

By Parts (LIATE):

$$\int u dv = uv - \int v du$$

Definite Integral (FTC):

$$\int_a^b f(x) dx = F(b) - F(a)$$

MVT:

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

For feedback or to request additional topics:

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